

FX Simulation Model for PFE

Model Assumptions & Limitations Validation $\rightarrow$ Potential Findings

Model Validation EMRM

MUFG

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Agenda

- 1 Model Description and Calibration
- 2 Model Assumptions
- 3 Model Limitations
- 4 Statistical Validation
- 5 Conclusions and Recommendations

Model Purpose

Simulate FX rate evolution for Potential Future Exposure (PFE)

Model Framework:

- Geometric Brownian Motion (GBM)
- Log-normal distribution
- Constant volatility
- Independent currency pairs

Key Features:

- Risk-neutral drift (CIP)
- Historical volatility calibration
- 3-year lookback window
- 5-day return methodology

Stochastic Process

Each FX rate $Y_k^{FX}(t)$ follows:

$$dX_k^{FX}(t) = \sigma_k dW_k(t)$$

$$Y_k^{FX}(t) = g_k^{FX}(t) \exp\left(X_k^{FX}(t) - \frac{1}{2}\sigma_k^2 t\right)$$

where:

- $W_k(t)$: Standard Brownian motion
- σ_k : Constant volatility parameter
- $g_k^{FX}(t) = Y_k^{FX}(0) \frac{DF_f(0,t)}{DF_d(0,t)}$: Mean function (covered interest parity)

Drift Calibration (Risk-Neutral $\mathbb{Q}$):

$$g_k^{FX}(t) = Y_k^{FX}(0) \frac{DF_f(0, t)}{DF_d(0, t)}$$

- Derived from interest rate curves
- Ensures no-arbitrage at $t = 0$
- Consistent with forward prices

Volatility Calibration (Historical $\mathbb{P}$):

$$\sigma_k = \text{StdDev} \left[\ln \left(\frac{Y_k(t + 5d)}{Y_k(t)} \right) \right] \times \frac{1}{\sqrt{5}}$$

- 3-year historical window
- 5-day log returns
- Scaled to daily frequency

Mixed-Measure Approach

Drift from $\mathbb{Q}$ + Volatility from $\mathbb{P}$ = **Theoretical Inconsistency**

Calibration Inputs

Required Data

For each FX pair k :

- ① **Historical Data:** 3-year time series of daily FX spot rates
 - Used to compute 5-day log returns
 - Calculate standard deviation
- ② **Interest Rate Curves:** Current discount factors
 - Domestic curve: $DF_d(0, t)$
 - Foreign curve: $DF_f(0, t)$
- ③ **Current Spot:** $Y_k^{FX}(0)$
 - Market observable at calibration time

Core Assumptions (1/3)

Assumption 1: Mixed-Measure Calibration

Drift: Risk-neutral (from CIP)

Volatility: Historical (from realized returns)

⇒ Assumes $\sigma^{\mathbb{P}} \approx \sigma^{\mathbb{Q}}$ (ignores vol risk premium)

Assumption 2: Log-Normal Distribution

FX rates follow log-normal distribution implying:

- Rates strictly positive
- Log-returns normally distributed
- No fat tails (thin-tailed)

⇒ Underestimates extreme events

Core Assumptions (2/3)

Assumption 3: Constant Volatility

Volatility σ_k is time-invariant and state-independent

$$\text{Var}[r_{0,t}] = \sigma_k^2 t \quad (\text{linear in time})$$

⇒ Ignores volatility clustering, term structure, stochastic volatility

Assumption 4: Historical Calibration

- 5-day returns assumed i.i.d.
- 3-year window stationary
- Variance scales linearly: $\sigma_{\text{daily}} = \sigma_{5d}/\sqrt{5}$

⇒ smooths spikes, ignores regime changes

Core Assumptions (3/3)

Assumption 7: No Jumps

Pure diffusion, no discontinuous movements

⇒ Misses central bank interventions, crashes, news shocks

Assumptions Summary Table

Assumption	Model Specification	Reality
Mixed-measure	$\mathbb{Q}$ drift + $\mathbb{P}$ vol	Vol risk premium: $\sigma^{\mathbb{Q}} > \sigma^{\mathbb{P}}$
Log-normal	Normal log-returns	Fat tails, kurtosis $\approx 5-10$
Constant volatility	$\sigma(t, S) = \sigma$	Clustering, smile, term structure
Historical calibration	5-day, 3-year window	Market uses implied vol
Independence	$\rho_{ij} = 0$	Empirical: $\rho \approx 0.65$ (USD crosses)
Deterministic rates	$DF(0, t)$ frozen	Rates stochastic, correlated with FX
No jumps	Pure diffusion	Central bank interventions, crises

Limitation 1: Mixed-Measure Calibration

Fundamental Inconsistency

Model combines risk-neutral drift with historical volatility

Theoretical Issues:

- **Volatility Risk Premium Ignored:** Market implied vol exceeds realized vol by 2–4%
- **FX Risk Premium Omitted:** UIP fails empirically (carry trades profitable)
- **Measure Ambiguity:** Neither pure $\mathbb{Q}$ nor pure $\mathbb{P}$

Impact:

- × PFE underestimated by **8–20%**
- × Option values underpriced by 10–15%
- × Hedge ratios wrong by 15–30%

Limitation 2: Deterministic Interest Rates

Pathwise No-Arbitrage Failure

Frozen discount factors create arbitrage opportunities on individual paths

Problem: Conditional expectation incorrect

$$\text{Should be: } \mathbb{E}_s^{\mathbb{Q}}[S(t)] = S(s) \frac{DF_f(s, t)}{DF_d(s, t)}$$

$$\text{Model uses: } \mathbb{E}_s^{\mathbb{Q}}[S(t)] = S(0) \frac{DF_f(0, t)}{DF_d(0, t)} \quad \leftarrow \text{Wrong!}$$

Impact:

- Long-dated PFE error: **15–30%**
- Callable/Bermudan mispricing

Limitation 3: Constant Volatility

What the Model Misses:

① Volatility Clustering

- High vol $\rightarrow$ high vol
- Low vol $\rightarrow$ low vol
- Persistence: $\alpha + \beta \approx 0.95$

② Term Structure

- Short-term vol $>$ long-term
- Mean-reversion effects

③ Stochastic Volatility

- Vol itself is random
- Creates smile/skew

Quantitative Impact:

- PFE underestimation: **15–30%**
- Tail risk ($\text{VaR}_{99\%}$): **20–40%** error
- Path-dependent mispricing

Example:

- Calibrated: $\sigma = 8\%$
- Stress spike: $\sigma = 14\%$
- PFE error: **38%** underestimation

Limitation 4: No Correlation Structure

Independence Assumption Violated

Empirical FX pairs exhibit significant correlation

Empirical Correlations:

Pair	Correlation
EUR/USD $\times$ GBP/USD	0.65
EUR/USD $\times$ AUD/USD	0.55
USD/JPY $\times$ EUR/USD	-0.20

Consequences:

- Concentration risk missed
- Diversification ignored
- Triangular arbitrage violated
- Portfolio VaR error

Example

EUR/USD + GBP/USD portfolio: Model underestimates risk by **22%** due to ignored positive correlation

Limitations Summary

Limitation	Root Cause	Impact TbC	Affected
Mixed-measure	$\mathbb{Q} + \mathbb{P}$	8–20%	All metrics
Constant vol	Time-invariant	15–30%	Tail risk, PFE
Historical vol	Ignores implied	10–15%	Options
No correlation	Independent	15–30%	Portfolio
No jumps	Pure diffusion	20–40%	VaR, barriers
Deterministic rates	Frozen DF	20–40%	XCS
Pathwise arbitrage	Conditional mismatch	15–35%	PFE
Path-dependent	Compound	25–50%	Barriers, TARFs

Test 1: Ljung-Box (Volatility Clustering)

Purpose: Test for autocorrelation in squared returns (volatility clustering)

Null Hypothesis: $H_0: \text{Corr}(r_t^2, r_{t-k}^2) = 0$ for all k

Test Statistic:

$$Q(K) = T(T+2) \sum_{k=1}^K \frac{\hat{\rho}_k^2}{T-k} \sim \chi^2(K)$$

Results (EUR/USD):

Lag	Q-stat	p-value
1	24.8	< 0.001
5	87.3	< 0.001
10	142.5	< 0.001
20	198.2	< 0.001

Conclusion

REJECT H_0 — Strong evidence of volatility clustering. Constant volatility assumption **VIOLATED**.

Test 2: ARCH-LM (Heteroskedasticity)

Purpose: Test specifically for ARCH effects (volatility clustering)

Null Hypothesis: H_0 : No ARCH effects (constant volatility)

Procedure:

- 1 Regress r_t on lags
- 2 Get residuals $\hat{\epsilon}_t$
- 3 Regress $\hat{\epsilon}_t^2$ on lags
- 4 Compute LM statistic

Results (EUR/USD):

- LM Statistic: 127.4
- p-value: < 0.001
- F-statistic: 13.2
- F p-value: < 0.001

Conclusion

REJECT H_0 — Overwhelming evidence of ARCH effects. Constant volatility model **INADEQUATE**.

Test 3: GARCH vs Constant Volatility

Purpose: Compare model fit between constant vol and GARCH(1,1)

Model Comparison:

Metric	Const	GARCH
Log-Lik	-2,845	-2,672
AIC	5,692	5,352
BIC	5,697	5,367

GARCH Parameters:

- $\alpha = 0.082$ (shock)
- $\beta = 0.905$ (persistence)
- $\alpha + \beta = 0.987$

Likelihood Ratio Test:

- LR stat: 346
- p-value: $< 10^{-74}$
- Improvement: 173 log-lik points

Out-of-Sample Forecast:

- Const Vol RMSE: 0.000847
- GARCH RMSE: 0.000621
- Improvement: **26.7%**

Conclusion

GARCH(1,1) **significantly superior** to constant volatility

Test 4: Volatility Term Structure

Purpose: Test if implied volatility varies across maturities

Market Data (EUR/USD):

Maturity	Vol	vs 3M
1 Week	11.8%	+24%
1 Month	10.5%	+11%
3 Months	9.5%	–
6 Months	9.3%	–2%
1 Year	9.9%	+4%
2 Years	10.7%	+13%

ANOVA Test:

- $H_0: \sigma(T_1) = \dots = \sigma(T_6)$
- F-statistic: 47.3
- p-value: < 0.001

Result

REJECT H_0

Term structure highly significant

Implication: Constant volatility across all maturities is **empirically rejected**

Statistical Evidence Summary

Test	Null Hypothesis	Result	Conclusion
Ljung-Box	No autocorr in r^2	$Q = 198, p < 0.001$	Reject
ARCH-LM	No ARCH effects	$LM = 127, p < 0.001$	Reject
GARCH comparison	Models equivalent	$LR = 346, p < 10^{-74}$	GARCH superior
Forecast RMSE	Equal accuracy	27% improvement	GARCH better
Term Structure	Flat vol surface	$F = 47, p < 0.001$	Reject

Overall Assessment (TbC)

All statistical tests **decisively reject** the constant volatility assumption. Model underestimates PFE by **15–25%** and fails regulatory standards.

① Fundamental Issue: Mixed-Measure Calibration

- Combines $\mathbb{Q}$ drift with $\mathbb{P}$ volatility
- Systematic underestimation: **8–20%**
- Affects all risk metrics

② Constant Volatility Fails Empirically

- All statistical tests reject assumption
- Misses clustering, term structure, stochastic dynamics

③ Missing Features Compound Errors

- No correlation: portfolio error
- No jumps: tail risk error
- Deterministic rates: XCS pricing error
- Path-dependent: mispricing

Material Impact by Portfolio Segment

High Impact Segments

- **Long-dated exposures** (> 2 years): Deterministic rates + constant vol
- **Barrier-heavy portfolios**: Jump risk + smile omission
- **Cross-currency swaps**: Pathwise no-arbitrage + rate-FX correlation
- **Multi-currency portfolios**: Correlation structure critical

Acceptable for

- Short-dated ($< 6M$) vanilla forwards
- Single-currency, linear exposures
- Low-volatility major pairs (EUR/USD, USD/JPY) in calm markets

Recommendations

Priority/Option 1

- ① **Volatility Adjustment:** Apply 10–15% uplift to historical vol

$$\sigma_{\text{adjusted}} = \sigma_{\text{historical}} \times 1.12$$

- ② **Diagnostic Testing:** Implement Ljung-Box, ARCH-LM tests

Priority/Option 2

- ① **Hybrid Volatility:** Combine historical + implied (60/40 weight)
- ② **Correlation Matrix:** Implement Cholesky decomposition
- ③ **Predictor-Corrector:** Periodic drift updates for long-dated

Recommendations(continued)

Priority/Option 3: Long-Term (Strategic, 12-24 months)

- ① **Stochastic Volatility:** Implement Heston model
 - Captures clustering, smile, fat tails
 - Industry standard for FX options
- ② **Jump-Diffusion:** Add Merton or Kou jumps
 - Captures extreme events
 - Critical for barrier portfolios
- ③ **Stochastic Rates:** Integrate Hull-White or LMM
 - Pathwise no-arbitrage maintained
 - Correct rate-FX linkages

Goal

Transition to market-consistent, multi-asset framework (as per XVA)

Summary of Validation Findings

Model Strengths:

- ✓ Computational efficiency
- ✓ CIP drift ensures no-arbitrage at $t = 0$
- ✓ Tractable, closed-form moments
- ✓ Suitable for short-dated vanillas

Critical Weaknesses:

- ✗ Mixed-measure calibration
- ✗ Constant volatility fails tests
- ✗ No correlation structure
- ✗ No jump dynamics
- ✗ Pathwise arbitrage

Overall Assessment

Model is **adequate for basic PFE** but requires enhancements for:

- Barrier/path-dependent portfolios
- Long-dated cross-currency exposures
- High-volatility/EM currencies

SDE Specification:

$$dX_k^{FX}(t) = \sigma_k dW_k(t)$$

$$Y_k^{FX}(t) = g_k^{FX}(t) \exp \left(X_k^{FX}(t) - \frac{1}{2} \sigma_k^2 t \right)$$

$$g_k^{FX}(t) = Y_k^{FX}(0) \frac{DF_f(0, t)}{DF_d(0, t)}$$

Log-Return Distribution:

$$R_{s,t} := \ln \left(\frac{Y_k^{FX}(t)}{Y_k^{FX}(s)} \right) \sim \mathcal{N} \left((r_d - r_f - \frac{1}{2}\sigma_k^2)(t-s), \sigma_k^2(t-s) \right)$$

Monte Carlo Implementation:

- ① Generate $Z \sim \mathcal{N}(0, 1)$
- ② Update: $X(t + \Delta t) = X(t) + \sigma\sqrt{\Delta t} \cdot Z$
- ③ Compute: $Y(t + \Delta t) = g(t + \Delta t) \exp(X(t + \Delta t) - \frac{1}{2}\sigma^2(t + \Delta t))$

Backup: GARCH Specification

GARCH(1,1) Model:

$$r_t = \mu + \sigma_t \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, 1)$$
$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$$

Interpretation:

- ω : Long-run baseline variance
- α : Reaction to recent shocks (ARCH effect)
- β : Persistence of volatility (GARCH effect)
- $\alpha + \beta$: Overall persistence (near 1 = long memory)

Empirical Values (EUR/USD):

- $\omega = 0.0008$, $\alpha = 0.082$, $\beta = 0.905$
- Persistence: $\alpha + \beta = 0.987$ (very close to 1)
- Half-life: ≈ 53 days

Key Literature:

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